

# Semantics-based Connectivity Graph for Indoor Pathfinding Powered by IFC-Graph

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**Abstract:** Pathfinding is important for many indoor location-based applications such as facility management and indoor navigation. Conventional approaches for constructing indoor networks for pathfinding are mostly based on geometry processing, which can be computing-intensive and time-consuming, especially for buildings with complex irregular shapes. In addition, the generated geometric networks often lack semantics. This paper proposes a new semantics-based approach for constructing navigable indoor networks. This approach uses the rich semantic information from Industry Foundation Classes (IFC) in the form of IFC-Graph (i.e., a graph-based representation of building information developed in a prior study) so as to meet the needs of indoor pathfinding. Real-world building models are used to validate the proposed approach. The results show that 1) the semantics-based approach has advantages in object-to-object pathfinding such as space-to-space, space-to-exit, and space-to-facility, 2) the use of graph technology makes it flexible to add semantic information to the network to better support pathfinding applications. The semantics-based approach can fit into digital twin technology and integrate with augmented reality.

**Keywords:** Indoor Navigation and Pathfinding; Labelled Property Graph (LPG); IFC-Graph; Building Information Modeling (BIM); Industry Foundation Classes (IFC); Geographic Information System (GIS); Digital Twin (DT)

## 1 Introduction

This paper addresses the problem of the efficient construction of semantically rich connectivity graph for indoor navigation by utilising IFC-Graph, i.e., a new graph-based representation of Industry Foundation Classes (IFC) developed in a prior study by Zhu et al. [1]. Connectivity graphs of buildings show how spaces (e.g., rooms and corridors) in buildings are connected with each other. They are used in many areas, such as automation, architecture, and facility management [2,3]. The common applications of connectivity graphs include building design [4], indoor navigation and pathfinding/tracking [5], emergency response and evacuation [6], and other indoor location based services (LBS). Efficient indoor pathfinding is essential to improving the experience of built environment users, such as residents, visitors, and facility managers, as people spend about 90% of their time in indoor spaces [7].

Indoor pathfinding largely relies on indoor networks that indicate the connectivity of indoor spaces. Indoor networks are also referred to by other names, such as navigation network graph [8], indoor graph network [9], route graph [10], geometric network model [11], or indoor spatial model [12,13]. Despite the variety in the name and the structure of networks, connectivity is the shared feature. The automatic generation of navigable connectivity graphs from geometric information, such as 2D floor plans, has been explored [10]. Various methods have been developed, such as the visibility graph, straight skeleton, and generalised Voronoi graph. These methods mainly use geometric information to generate connectivity graphs, which is computing intensive and time-consuming [3,14]. According to Teo and Cho [9], it is complex and time-consuming to construct a geometric

network model (GNM), especially when the networks need additional semantic information. Lastly, these geometry-based navigation networks, without additional semantic information, have difficulty in supporting many practical applications, such as knowing ‘the locations of the nearest fire suppression terminal’ and ‘the number of available exits’ are crucial.

The importance of semantic information, i.e., meaning of spaces and their attributes and properties, as well as relationships in building information models (BIM) in constructing indoor navigation network has been increasingly acknowledged [9,15]. Fichtner et al. [8] claimed that pathfinding can profit from the rich semantics of its surroundings. BIM models in IFC format provides not only rich geometric information about the built environment but also abundant non-geometric semantic information, such as attributes, properties, and relationships. Several studies have explored the use of semantic information in the construction of indoor navigation network, such as Zhou et al. [3], Lin and Lin [12], Xu et al. [16], Zhao et al. [17] and Teo and Cho [9]. However, these studies have defined semantics from a navigation perspective. The semantic information used in these studies is limited to the type of entities (i.e., room, corridor) and some attributes of building elements, which directly serve the navigation tasks. Most of the studies classify the building elements into ‘navigable’ and ‘non-navigable’ space. For example, Liu defined semantics with respect to horizontal and vertical options for movement in addition to the use of spaces for navigation (e.g., vertical/horizontal connector, door, doorway, etc) [14]. IFC has defined far richer semantics than that, i.e., in IFC4 there are 776 entities defined for various objects, relationships, and properties [18].

Therefore, in the context that geometry-based approaches generate indoor networks that lack semantics in a computing-intensive and time-consuming way and that the semantics-based approach has not been sufficiently established, this paper aims to establish a semantics-based approach for indoor pathfinding by using semantic information from IFC in the form of IFC-Graph [1]. The remainder of this paper is organised as follows. Section 2 introduces the related work on indoor pathfinding and navigation. Section 3 presents the proposed semantics-based indoor connectivity graph. Section 4 is about the validation of the proposed approach. Discussions are presented in Section 5, and Section 6 concludes the paper.

## 2 Research background and related works

### 2.1 Indoor pathfinding and indoor navigation models

Indoor pathfinding is closely related to the broader topic of indoor navigation that has five related topics, including a) indoor localisation and positioning, b) indoor navigation model, c) algorithms for indoor pathfinding, d) human spatial cognition and wayfinding, e) indoor guidance instruction (e.g., verbal or graphical) [14]. Among these five topics, all the other topics depend on the navigation model, which represents the semantics, topology, geometry, and other context information of indoor environment.

Indoor navigation models have been developed in many areas, such as mobile robot mapping [19,20], geographic information system (GIS) [21], and ubiquitous computing [22,23]. The literature has presented different ways to group these navigation models, but in general they can be divided into three basic groups, i.e., metric-based, topology-based, and their combination [10].

1) Metric-based: The metric-based navigation models use exact coordinates in a global coordinate system. Models of this kind include the 2D grid-based models and 3D voxel-based models [24]. A grid-based model tessellates the 2D indoor space with a finite number of connecting but non-overlapping cells [3], either regular cells (e.g., square or hexagon) or irregular cells, while a voxel-based model represents the 3D space with non-overlapping voxels. Each cell or voxel represents either a passable or unpassable region that is differentiated by its states, e.g., 0 for unpassable and 1 for passable. The spaces represented by the grid-based or voxel-based models are spatially continuous (either 2D or 3D) and are mainly used in gaming and robotics-based autonomous navigation and mapping [25]. To create such navigation models, a common workflow is a) the tessellation or voxelization of space, b) the space mapping between cell/voxel and building geometry, and c) generation of indoor navigation maps.

2) Topology-based, also referred to as graph-based models: The topology-based models represent the indoor spaces with nodes and the connectivity between them with edges. Poincare duality is often used to establish relationship between spaces and topological (network) representations [26]. Topology-based models are used to denote relationships between passable spaces which are appropriate for navigation. The relationships can represent connectivity, accessibility, adjacency, and inclusion. Models of this kind generally qualitatively indicate the relationship, e.g., connectivity. Such graph-based models are also called logical graphs and are used in conceptual analysis [14].

3) Network (combining geometry and topology): To use the advantages of the above two types of navigation models, a third type of navigation model has been created, referred to as network or route graph, which combines geometry from metric-based models with connectivity from topology-based models [10]. Similar to topological models, a network uses nodes to indicate objects/spaces (rooms, doors) and uses edges to indicate their connectivity. The nodes in networks also contain location information, which can be used to calculate distance between objects. Edges can also be described as curves, representing the real path to follow. Most of the current indoor navigation models are networks, such as geometric network model (GNM), improved geometric network model (IGNM) [15], multi-purpose geometric network model, lattice topology data model (LTDM) [27], visibility graph (VG) [28,29], straight skeleton from straight medial axis transformation (S-MAT) [30,31], generalised Voronoi graph (GVG), and IndoorGML models constructed using Poincare duality [26,32].

A summary of the advantages and disadvantages of these indoor navigation models are provided in Table 1. The metric-based models suffer from performance problems related to path searching and model storage, especially for large navigation maps derived from the use of a fine cell/voxel size to tessellate space; the topology-based models suffer from the lack of geometric information; and the various route graphs suffer from different problems. For example, the paths from VG usually do not follow the shape of building space [14], GVG may generate excessive nodes and paths in irregular zones, the S-MAT algorithm can get stuck at concave vertices of a simple polygon, and the modified MAT algorithm does not accurately represent the centreline of the space [33]. These geometry-based methods for navigation graph construction are less efficient for complicated built environment, and their computational costs increase significantly with the increment of grids, nodes, and edges. It is still a challenge for them to deal with irregular floor layouts. That being said, the selection of the indoor navigation model should be based on the needs of projects. A more comprehensive review about these navigation models is presented in Pawel et al. [26].

Table 1 Common indoor navigation models and their advantages and disadvantages.

|   | Category       | Type                                                                                                                                                               | Advantages                                                                        | Disadvantages                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Metric-based   | <ul style="list-style-type: none"> <li>Grid-based</li> <li>Voxel-based</li> </ul>                                                                                  | <ul style="list-style-type: none"> <li>Spatially continuous</li> </ul>            | <ul style="list-style-type: none"> <li>Computing-intensive.</li> <li>The grid/voxel size affects storage and pathfinding performance.</li> <li>Object boundaries are jagged.</li> </ul>                                                                                                                                                                                                                                                                                                                        |
| 2 | Topology-based |                                                                                                                                                                    | <ul style="list-style-type: none"> <li>Compact structure</li> </ul>               | <ul style="list-style-type: none"> <li>No geometric information for navigation guidance.</li> <li>Conceptual analysis only.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                         |
| 3 | Network        | <ul style="list-style-type: none"> <li>Visibility Graph</li> <li>Straight skeleton from S-MAT</li> <li>Generalised Voronoi Graph</li> <li>Triangulation</li> </ul> | <ul style="list-style-type: none"> <li>Combining geometry and topology</li> </ul> | <ul style="list-style-type: none"> <li>Difficult in defining start and end point, not suitable for indoor positioning.</li> <li>The path does not follow the shape of the building spaces.</li> <li>The number of edges increase rapidly.</li> <li>The increased number of edges lead to complicated interpretation of route.</li> <li>Weird network for corridors may be created.</li> <li>Does not work for large arbitrary shaped spaces.</li> <li>Excessive nodes and paths in irregular zones.</li> </ul> |

## 2.2 BIM-driven indoor pathfinding

The use of BIM in indoor pathfinding is a response to the problem within early 3D models from CAD (Computer Aided Design) that cannot provide 'appropriate and applicable' representations of building geometry and detailed semantics of indoor environment [9,34]. BIM is a technology that creates digital models for built assets that can be used in whole lifecycle management [35]. Building models from BIM contain rich geometric, semantic, and topological information and can serve as a good source of information for the construction of indoor navigation models. Wong et al. used IFC data for constructing visibility graph in their CORE system in response to fire emergency [5,6,36]. Chen et al. used geometric and semantic information from IFC to construct voxel-based indoor 3D map [37]. Lin used IFC data to create high accuracy paths for winder, spiral, and straight stairs, which can be used to improve the overall accuracy of indoor geometric network [38]. Zhou et al. created grid-based map from IFC data, which was further converted into a topological map for indoor pathfinding [3]. Teo and Cho used IFC data to create their multi-purpose geometric network model (MGNM) [9].

## 2.3 Issues in BIM-driven indoor pathfinding

The construction of indoor navigation network is still driven by geometry. Previous studies mainly used geometric information, e.g., 2D floor plan, to derive topological information (i.e., adjacency graph and connectivity graph) for indoor pathfinding. These geometry-based topological networks, due to the lack of semantics, have difficulty in answering questions like 'where is the nearest fire suppression terminal?' and 'how many exits are available, and which is the nearest?'. There are efforts to embed semantics information into the geometry-based topological networks, but embedding and updating semantic information in geometric networks is time-consuming [9], even though the semantic information used is already quite limited. The most used semantic information is limited to the name of entities and a small number of attributes, such as the unique identifier (ID) [39], the material [40], the 'FireExit' attribute of doors [6], and the 'PubliclyAccessible' for spaces [24]. In short, the conventional geometry-based approach (i.e., geometry supplemented with semantics) has efficiency problems in network generation and updating, as well as functional problems when semantics need be considered.

The semantics-based approach (i.e., semantics supplemented with geometry) has been less investigated. The only attempt that can be found in literature is from Ismail et al. [41], who used IFC to construct a semantics-only labelled property graph and used the graph in indoor pathfinding. This very initial attempt experienced several problems related to the navigation data model defined, the use of semantics, the query for the shortest path, and the visualisation of the path for navigation. In a related study by Khalili and Chua [40], even though they attempted to establish a graph data model (GDM) through a topology-driven approach, they ended up with a geometry-based approach by using geometric information from boundary representations (B-reps), and the derived topology graphs have accuracy issues. The lack of study on the semantics-based approach can probably be attributed to the fact that previous studies believe that topology graphs (without geometry) can only be used in conceptual analysis, not 'real' pathfinding. Furthermore, the conventional way for generating topology graphs is using the geometry-based approach, which is time-consuming [7,42].

## 3 Semantics-based indoor navigation graph powered by IFC-Graph

Motivated by the issues identified in literature and latest advances in building information technology such as IFC-Graph [1], this paper aims to explore and enable semantics-based indoor pathfinding by developing a semantics-based connectivity graph. IFC-Graph is a semantic network that represents the rich semantic, geometric, and topological information in IFC. The semantic information in IFC refers to attributes and properties of building and spatial elements; the geometric information refers to the shape, size, and location, and the topological information is derived from relationships in IFC. Relationship entities in IFC are used to connect objects and indicate relation, for example spatial aggregation, spatial containment, and element connectivity.

To achieve the research aim, this paper mainly addressed the following challenges: 1) investigating the IFC standard to assess its capability in supporting indoor pathfinding, 2) exploring how the information in IFC data can be efficiently extracted to construct the semantics-based connectivity graph, 3) exploring how the semantics-based connectivity graph can be used in indoor pathfinding. Figure 1 presents the detailed tasks in the workflow, including 1) establishing an indoor navigation data model, i.e., the data model behind the connectivity graph, 2) establishing IFC-Graph from the IFC STEP Physical File (IFC-SPF), 3) establishing horizontal connectivity graph for individual storeys, 4) obtaining geometries for vertical connecting elements (e.g., stairs) and spaces, 5) establishing vertical links between storeys, 6) establishing building-wide connectivity graph, and 7) performing indoor pathfinding.

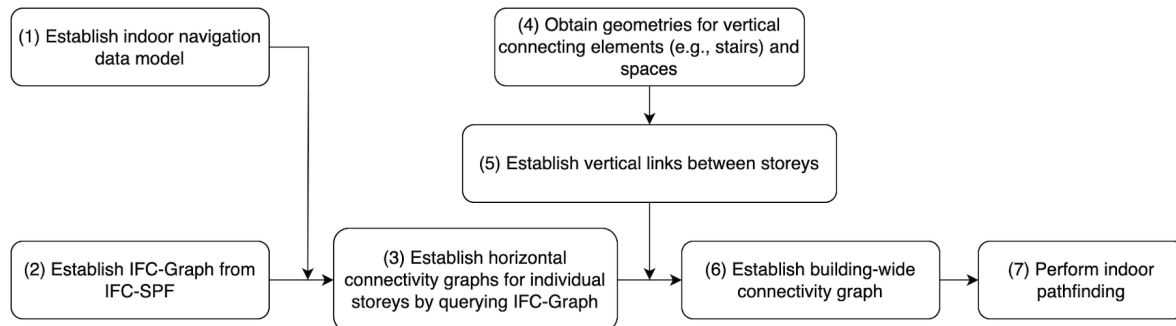


Figure 1 Workflow for constructing semantics-based connectivity graph for indoor pathfinding.

### 3.1 IFC-based indoor navigation data model

An indoor navigation data model is an abstract description of the indoor environment for navigation. There are general data models for representing indoor environments, such as CityGML and IFC, but they are not specifically for indoor pathfinding. Other navigation data models have been developed for geometric networks, such as the BIM-Oriented Indoor Data Model (BO-IDM) [34], the Indoor Navigation Space Model (INSM) [42], and IndoorGML [43]. This paper created a new data model for the semantics-based approach based on the IFC data model. The new navigation data model considers both the horizontal connectivity and the vertical connectivity. Horizontal connectivity is for building elements and spaces in the same storey, while vertical connectivity is for those on different storeys.

#### 3.1.1 Horizontal connectivity

Previous studies used geometric information to derive horizontal connectivity, without recognising that such information is already defined in IFC in the form of relationships, such as `IfcRelSpaceBoundary` and `IfcRelContainedInSpatialStructure`. The most important relationship entity for establishing horizontal connectivity is `IfcRelSpaceBoundary`, which connects spaces and their boundaries [44], as presented in Figure 2. The spaces (room, corridor, and others) constitute the indoor space of a building. The common boundaries for spaces are physical building elements (such as walls, slabs, doors, windows, and columns) or the virtual element (i.e., space/room separator or `IfcVirtualElement` in IFC). The space (`IfcSpace`), boundary relationship (`IfcRelSpaceBoundary`), and the boundary elements (`IfcElement`) form the basis of the horizontal connectivity graph. However, not every boundary element can be used to connect spaces, such as slabs or walls. Elements that are used to connect spaces are referred to as connecting elements in this paper, such as doors, windows, and openings. Other studies may use a different term to refer to the connecting elements, such as ‘portals’ used by Teo and Cho [9].



Figure 2 `IfcRelSpaceBoundary` entity for establishing horizontal connections.

### 3.1.2 Vertical connectivity

While IfcRelSpaceBoundary supports the establishment of horizontal connectivity, there is a problem in the vertical direction, i.e., IFC has not specified any relationships for vertical storey-to-storey connection [9,37,42]. IfcRelSpaceBoundary might be able to provide such semantics, as IFC defines that any 'IfcElement' can be involved as a boundary [44], but this use has not been adopted by the industry. The lack of vertical connection impedes the creation of a building-wide connectivity graph. Therefore, in this paper, a new relationship entity has been defined, i.e., IfcRelConnectsSpace (see Figure 3). This new relationship entity defines how vertical connections can be established by linking vertical connecting elements (such as stair, ramp, lift, and escalator) with spaces on different storeys. Overall, the space, connecting elements (both horizontal and vertical), IfcRelSpaceBoundary, and IfcRelConnectsSpace form the basis of the indoor navigation data model.



Figure 3 New IfcRelConnectsSpace entity for establishing vertical connections.

## 3.2 IFC-Graph supporting the indoor navigation data model

To implement the indoor navigation data model and construct the connectivity graph, information from IFC data is extracted. To facilitate this process, a graph-based representation of IFC data, IFC-Graph, was used. IFC-Graph is a graph-based representation of IFC data using labelled property graph (LPG), which makes it easier to access and query interconnected building information, especially the semantic relationship information [1,45,46]. Through pattern-based graph query, information about buildings can be easily revealed and queried.

### 3.2.1 IFC-Graph generation

In this study, the method presented in Zhu et al. [1] was used to generate IFC-Graph from IFC-SPF. The generation of IFC-Graph involves two steps: concept mapping and IFC-SPF conversion. 1) In concept mapping, the concepts in IFC (i.e., entities, attributes) are mapped to concepts in property graph (i.e., nodes, edges, and properties). To be specific, entities (including relationships) are mapped to nodes, and attributes of entities are mapped either to edges or to properties depending on the type of attribute value. 2) Based on the established mapping, instances in an IFC-SPF are traversed and converted into nodes. For attributes of instances, if the value of an attribute is a string or a number, the attribute will be converted into a property; if the value of an attribute is another instance, a node will be created for the referenced instance, and an edge will be created to connect these two nodes. After traversing each instance, the final IFC-Graph can be established. For more details about IFC-Graph generation, please refer to Zhu et al. [1]. An example of IFC-Graph is presented in Figure 4 using the house model from Karlsruhe Institute of Technology (KIT) [47].

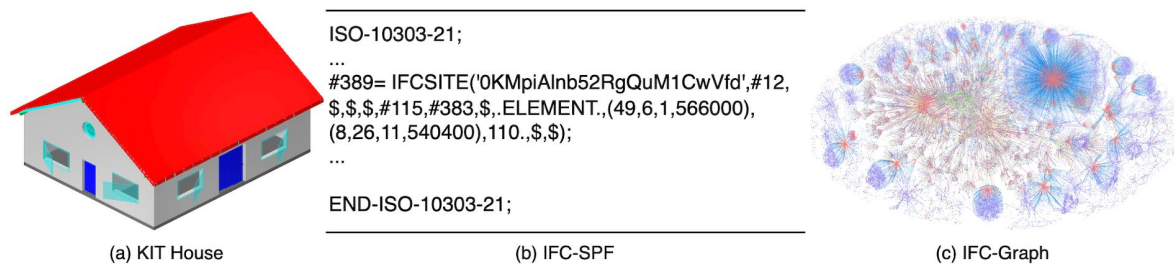


Figure 4 (a) House model in (b) IFC-SPF and (c) IFC-Graph.

### 3.2.2 IFC-Graph query

The IFC-Graph preserves all the information in the original IFC-SPF but in a graph form. To retrieve the needed building information from this graph structure, a graph query language (GQL)

[48] needs to be used, such as Cypher [49] or its open-source implementation openCypher [50]. In this study, the generated IFC-Graph is stored in neo4j. Therefore, its query language, Cypher, is used. The general steps for graph querying are 1) determining information needs, 2) designing the query pattern, and 3) executing the query. Cypher uses patterns to query information from graph. The graph patterns can be used to represent human knowledge about the indoor space [46] and used in building information querying and further knowledge discovery. As the basic elements of graph patterns, a node is indicated by '()', and the edge between two nodes is indicated by '--'. From these two basic elements, complex patterns can be defined for efficient information query. For example, the pattern '(:IfcSpace) -- (:IfcRelSpaceBoundary) -- (:IfcBuildingElement)' can be used to retrieve all spaces in a building model, as well as their bounding building elements. For more details about IFC-Graph query, please refer to Zhu et al. [46].

### 3.3 Generation of horizontal connectivity graph for individual storeys

While previous studies used geometric information to generate connectivity graphs, either manually or by using complicated algorithms, this study used IFC-Graph and graph query. The horizontal connectivity graph for each storey can be easily created by querying the IFC-Graph. In IFC, spaces are connected by doors or virtual elements. Virtual elements are used to identify 'notional' boundaries between spaces. This knowledge can be represented using the graph pattern in Listing 1 and used to construct the horizontal connectivity graph.

Listing 1 Graph query for constructing horizontal connectivity graph for individual storeys.

```
match p=(:IfcSpace) -- (:IfcRelSpaceBoundary) -- (n)
where n.name in ['IfcDoor', 'IfcVirtualElement']
return p
```

### 3.4 Establishing vertical connections for the building-wide connectivity graph

The horizontal connectivity graph for individual storeys can only be used in storey-level pathfinding. To enable building-level pathfinding, a building-wide connectivity graph for the whole building needs to be created. In this regard, storeys need to be connected via vertical connecting elements, such as stair, elevator, ramp, and escalator. IFC-SPFs, however, do not directly contain this information. Geometric information is needed to establish the vertical connection. In this paper, the geometries of vertical connecting elements and spaces were obtained from IFC, including shape, size, and location, and compared with each other through a 3D 'touch' analysis. This paper developed an algorithm to establish the vertical connection, as presented in Listing 2, which avoids computing-intensive 3D operations.

Listing 2 Algorithm for establishing vertical connection.

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Algorithm 1: Establish vertical connection via vertical connecting elements and spaces

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function compare\_vertical\_connecting\_elements\_with\_spaces (ifc\_file);

- 1 **Input:** An IFC-SPF file for a building  
**Output:** IDs of vertical connecting elements (VCEs) and connected spaces
- 2 **Get** all VCEs from ifc\_file
- 3 **For** each VCE in VCEs:
  - 4 **get** VCE's geometry and calculate its bounding box
  - 5 from the bounding box, **get** the centroid and the top elevation of the VCE
  - 6 [vce\_id, vce\_centroid\_x, vce\_centroid\_y, vce\_centroid\_z, vce\_top] → vce\_geo
- 7 **Get** all spaces from ifc\_file
- 8 **For** each space in spaces:
  - 9 **get** the space's geometry (shape, size, location)
  - 10 from the geometry, **get** the footprint of the space, and the minimal and maximal z\_value
  - 11 [space\_id, footprint, minimal z\_value, maximal z\_value] → space\_geo

Algorithm 1: Establish vertical connection via vertical connecting elements and spaces

---

```

12 For each VCE and each space:
13   If (the centroid of the VCE is within the footprint of the space) and (either vce_centroid_z or vce_top is within
    the range of [minimum z_value of space, maximum z_value of space]) then
14     return [vce_id, space_id]
15 end

```

---

In this algorithm, first, the geometries of vertical connecting elements (VCEs, e.g., stairs, ramps, etc.) and spaces are acquired from IFC-SPF and stored in two individual lists, `vce_geo` and `space_geo`; then, a 3D line was created for each vertical connecting element, from the centroid of the element to the top elevation; lastly, the 3D lines are compared with the shapes of spaces to see if they touch each other. If the 3D line of a VCE is within the footprint of a space and the minimum or maximum z-value of the VCE is between the minimum and maximum z-value of the space, the VCE is then assumed to ‘touch’ the space. The output is the connectivity between vertical connecting elements and spaces. It should be noted that, in IFC, stairs can be an assembly of its components, such as stair flight (`IfcStairFlight`), landing (`IfcSlab`), member (`IfcMember`), and railing (`IfcRailing`). In this case, stair flights are used to compute the vertical connectivity.

Once the vertical connectivity between storeys is established, it can be used to create a vertical connection subgraph, which can be used to construct the building-wide connectivity graph, together with the horizontal connectivity graph. The proposed approach can ensure that the building-wide connectivity graph includes all connected and accessible indoor spaces.

### 3.5 Indoor pathfinding

The building-wide connectivity graph can be used in indoor pathfinding by applying pathfinding algorithms, such as Dijkstra’s or A\* algorithms. In this paper, paths were calculated through graph queries from a graph database (Neo4j), which uses the Dijkstra’s algorithm. These algorithms can be performed via Neo4j Browser or via its application programming interface (API). The length of a path is defined as the number of the elements involved, and the shortest path is referred to as the least-element path, which is different from the shortest-distance path used by geometry-based approaches. In the study by Liu [14], this is referred to as the minimum navigable unit criterion. Listing 3 shows the query for finding the shortest path, in which ‘p’ is the variable for the queried path, ‘`shortestPath()`’ is the embedded function in neo4j for finding the shortest path, and ‘(start)-[\*]-(end)’ is the graph pattern indicating paths between the starting node (start) and the destination (end).

Listing 3 Query for finding the shortest (least elements) path.

```
match p = shortestPath((start)-[*]-(end)) return p
```

The queried paths can be used to indicate the sequence of spaces or connecting elements (e.g., doors and stairs) that are to be followed, but the sequence alone is abstract and not sufficient for indoor navigation, especially for novice users who are not familiar with the building; visual aids are also required for indoor wayfinding. In this regard, different from previous studies that used points and lines as visual aids, this paper uses the 3D (body) geometry of the involved elements. By doing this, instead of telling users which ‘point’ to go to or which ‘line’ to follow, this approach shows users which ‘space’ to go to. This approach also complies with the assumption by Stoffel et al. [51] that indoor pedestrians are less restricted and can travel freely between the interior boundaries of buildings.

### 3.6 Navigation guidance (visual aids)

To visualise the building elements in a path, geometric information needs to be retrieved from IFC data. Geometric information is mainly stored in two external attributes of IfcProduct, i.e., Representation and Placement. These two attributes point to IfcShapeRepresentation and IfcLocalPlacement, respectively. IfcShapeRepresentation stores the shape and size of geometry, while IfcLocalPlacement is about the location [52,53]. When visualising the shapes of elements in a path, the implicit parametric shapes in IFC represented as swept solid, constructive solid geometry (CSG), and clipping must be converted into explicit boundary representation (Brep) [54], and the coordinates need to be converted accordingly from the local coordinate system to the world coordinate system [9,55]. These operations can be performed by using computational geometry technologies [56,57]. Overall, the process for visualising the physical path from the graph path is as follows: 1) retrieving the shortest path from the building-wide connectivity graph, 2) identifying spaces and connecting elements in the graph path and retrieving their unique identifiers (IDs), 3) through these unique IDs, retrieving and constructing the geometry from the original IFC-SPF via geometry engine, such as Open CASCADE Technology (OCCT) for the visualisation of the physical path.

## 4 Validation and results

### 4.1 Data and methods

Two IFC models were used to validate the proposed approach (see Figure 5).

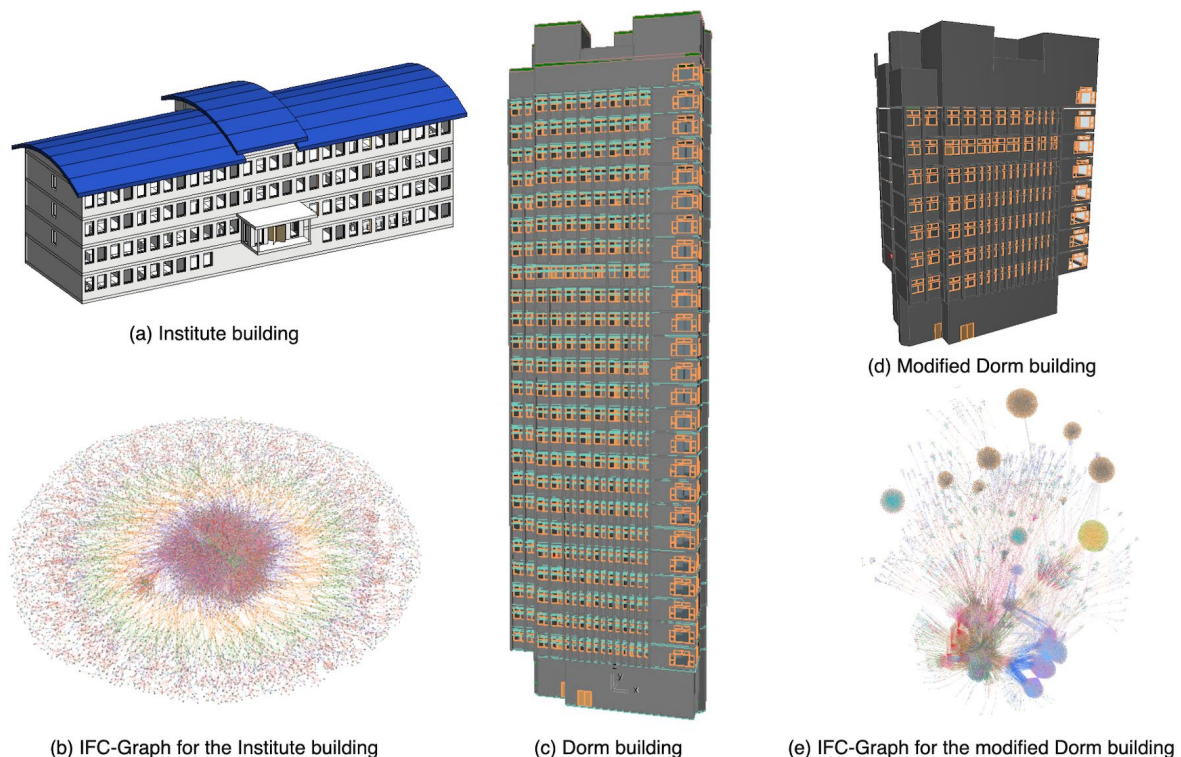


Figure 5 Building models, including (a) Institute building, (b) original Dorm building, and (c) modified Dorm building, as well as their IFC-Graphs.

The Institute model (Figure 5(a)) is for a building in KIT ([Google Earth](#)). The IFC-Graph for the Institute model has been generated and shown in Figure 5(b). The second Dorm model is for a building ([Google Earth](#)) in the University of Hong Kong (HKU). The original Dorm model (Figure 5(c)) has 27 storeys, with many of its storeys having the same floor plan. The duplicated floors were removed, resulting in the modified Dorm building (Figure 5(d)) that has 10 storeys with distinctive floor plans. The IFC-Graph for the modified Dorm building has been generated and is presented in Figure 5(e). An overview of these models is presented in Table 2.

Table 2 Overview of building models used.

| Model         | IFC-SPF   |                    |         | IFC-Graph |         |
|---------------|-----------|--------------------|---------|-----------|---------|
|               | File Size | Entities/Instances | Version | Nodes     | Edges   |
| Institute     | 10.9 MB   | 189/147712         | IFC4    | 194845    | 285504  |
| Dorm          | 40.1 MB   | 242/657433         | IFC4    | 675431    | 1377409 |
| Dorm-modified | 22.9 MB   | 211/327071         | IFC4    | 353749    | 619618  |

The following software tools were used in this study: IfcOpenShell for parsing IFC-SPF files, the neo4j graph database for storing the generated graphs (IFC-Graph and the connectivity graph), py2neo for interacting with the neo4j graph database, and OCCT for processing and visualising the geometry of building elements. The software packages used include Python 3.12.2, IfcOpenShell 0.7.0, py2neo 2021.2.4, OCCT 7.7.2, and Neo4j Desktop 1.4.15. The created neo4j database instances are in version 4.4.5. The computer used is equipped with an i7-13700K CPU, 32GB memory, and a NVIDIA GeForce RTX 3060 Ti graphic card.

#### 4.2 Institute model

The Institute model is a relatively small model, which is ideal for showing the details of the methods. This model has 5 storeys (see Figure 6), i.e., basement (-1), ground floor (0), first floor (1), second floor (2), and third floor (3). These 5 storeys have I-shaped floor plans and are connected by 4 vertical connecting elements, i.e., stairs.

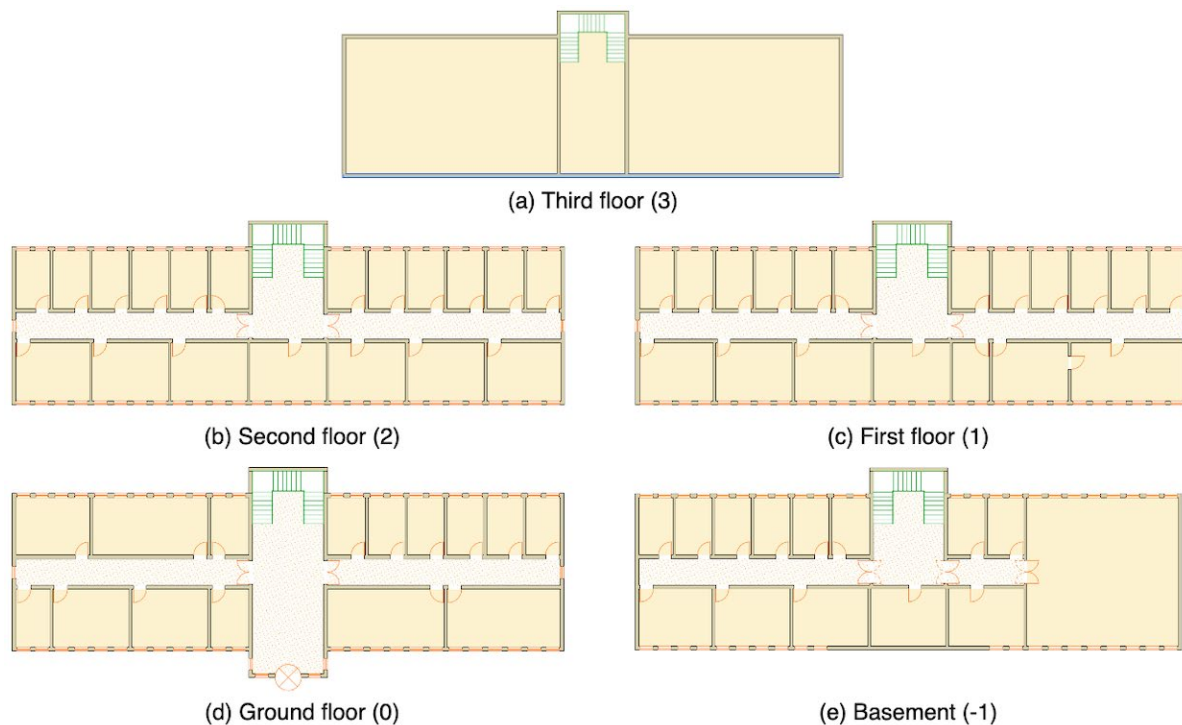


Figure 6 Floor plans for the Institute model.

Figure 7 presents the overall process of constructing the connectivity graph for the Institute model, including getting shapes of stairs and spaces (Figure 7(a)), linking stairs with spaces (Figure 7(b)), creating the stair-space link subgraph (Figure 7(c)), constructing the horizontal connectivity graph for individual storeys (Figure 7(d)), and creating the final building-wide connectivity graph (Figure 7(e)). The details of these steps are presented in the following sections.

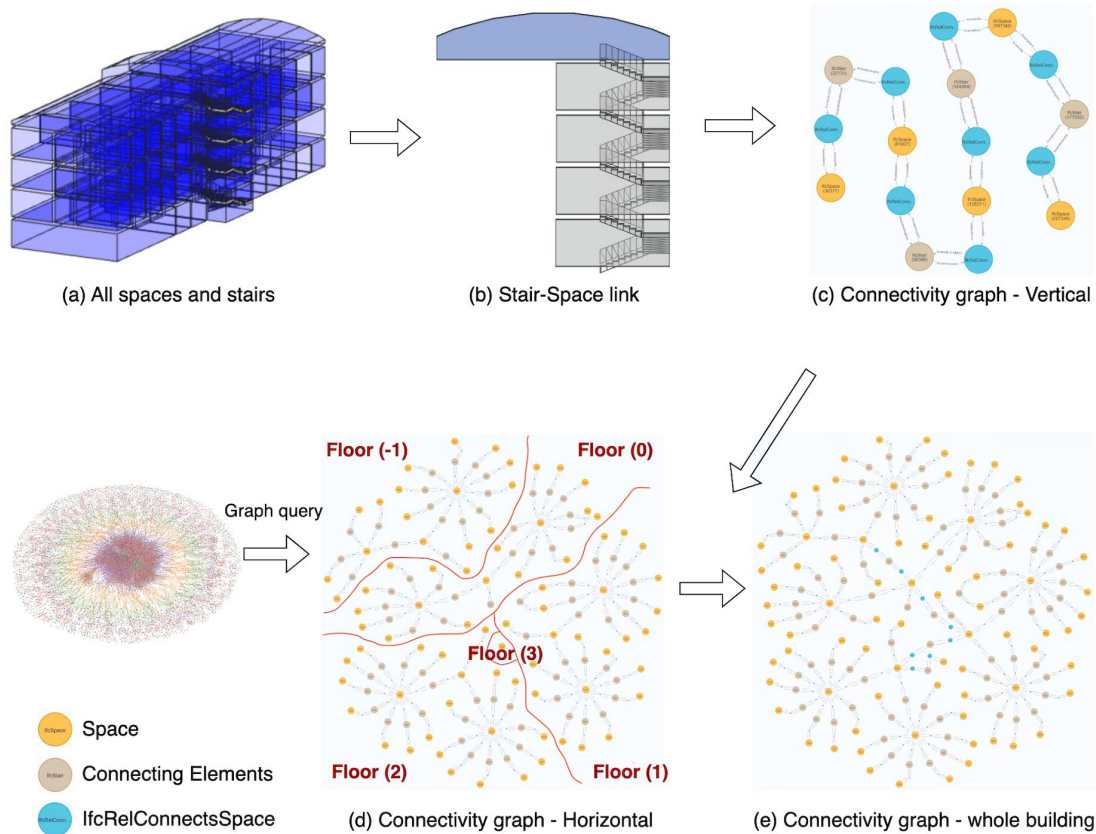


Figure 7 Construction of the connectivity graph for the Institute model.

#### 4.2.1 Horizontal connectivity graph for individual storeys

The horizontal connectivity graph for individual floors were obtained using the graph query in Listing 1, which finds all the spaces connected via either a door (IfcDoor) or a virtual boundary (IfcVirtualElement) through the IfcRelSpaceBoundary relationship. The query was completed in 2430 milliseconds (ms). Figure 8 shows an example of the connectivity graph for the ground storey.

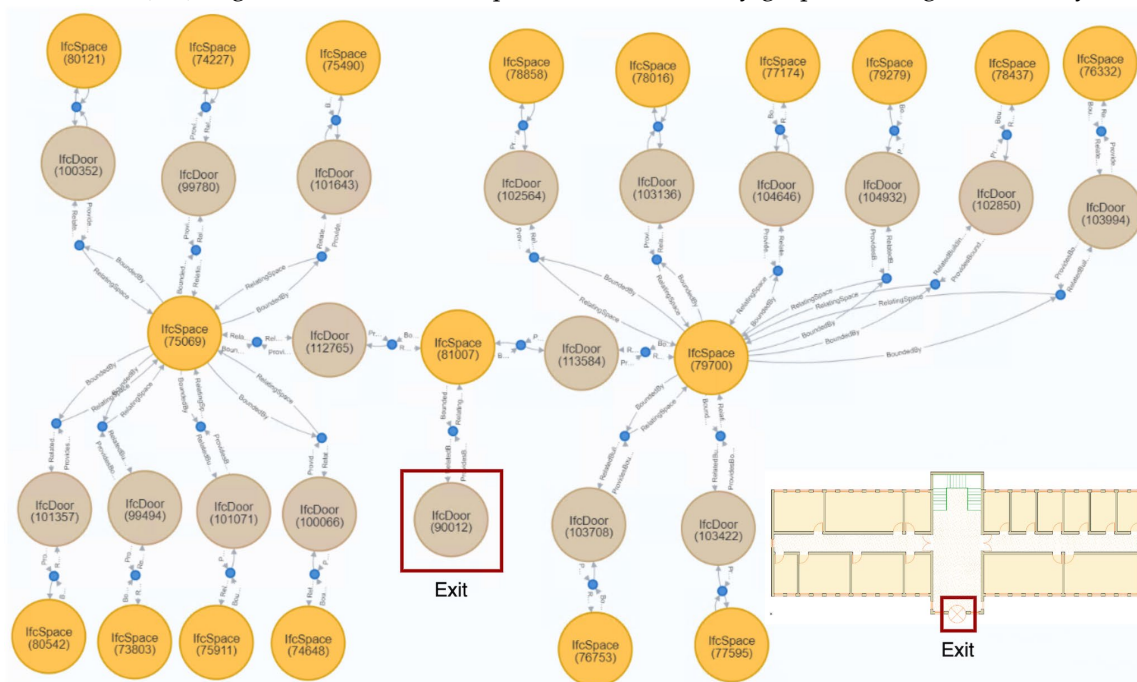


Figure 8 Connectivity graph for the ground floor.

It can be observed that the query has correctly retrieved the connectivity graph for the ground storey. It is also obvious that in this case the connectivity graph for the ground storey, as well as other storeys, presents a simple tree-like structure. In this simple structure, there is one and only one path between any two nodes. The query result (i.e., the horizontal connectivity graph) was saved by creating a new graph, which is to be combined with the vertical connectivity graph to form the building-wide connectivity graph.

#### 4.2.2 Vertical connections connecting storeys

By running the algorithm for identifying the relation between stairs and spaces, 8 links were identified, as listed in Table 3, which shows the unique identifiers of stairs and the connected spaces. Through these links and the newly defined IfcRelConnectsSpace relationship, a subgraph was created to represent these links, as presented in Figure 9, in which the blue dots are the newly defined IfcRelConnectsSpace instances. The connectivity graph for the whole building (Figure 7(e)) can be created by combining the horizontal connectivity graph with the newly created vertical connectivity graph.

Table 3 Stair-Space link in the Institute model.

| No | Stair_id | Space_id | No | Stair_id | Space_id |
|----|----------|----------|----|----------|----------|
| 1  | 22131    | 30377    | 5  | 124264   | 128271   |
| 2  | 22131    | 81007    | 6  | 124264   | 187346   |
| 3  | 58306    | 81007    | 7  | 177032   | 187346   |
| 4  | 58306    | 128271   | 8  | 177032   | 227340   |

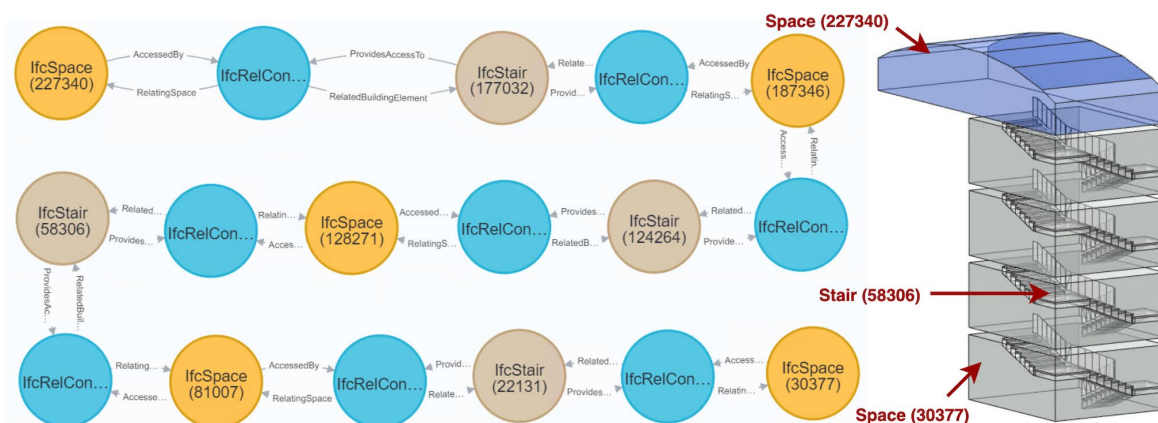


Figure 9 Vertical connectivity subgraph for the Institute model.

#### 4.2.3 Pathfinding using the building-wide connectivity graph

The building-wide connectivity graph can be used for indoor pathfinding. To demonstrate the semantics-based pathfinding, two objects were selected, i.e., a space (id:186061) and the exit door (id:90012). The graph query used to find the shortest path between these two objects is in Listing 4 and was completed in 42ms. The output of this query is a graph showing the path, which is 186061 (space) → 212897 (door) → 186903 (space) → 218941 (door) → 187346 (space) → 124264 (stair) → 128271 (space) → 58306 (stair) → 81007 (space) → 90012 (door), as presented in Figure 10(a). Using the unique identifiers, the shapes of involved elements were retrieved from IFC-SPF to construct the physical route as presented in Figure 10(b). This is the path that users can follow to leave the building. In this path, the landmarks, such as doors and stairs, are preserved for easier indoor wayfinding for users. As with geometry-based approaches that uses geometric points and lines as visual aid, this approach using 3D shapes of spaces and connecting elements (e.g., doors and stairs) is also intuitive. Nodes in the graph path can be labelled by any attributes defined in IFC, such as name or long name as presented in Figure 10 (c).

Listing 4 Query for finding the shortest path between space (id:186061) and door (id:90012).

```
match p = shortestPath((n1{id:186061})-[*]-(n2{id:90012})) return p
```

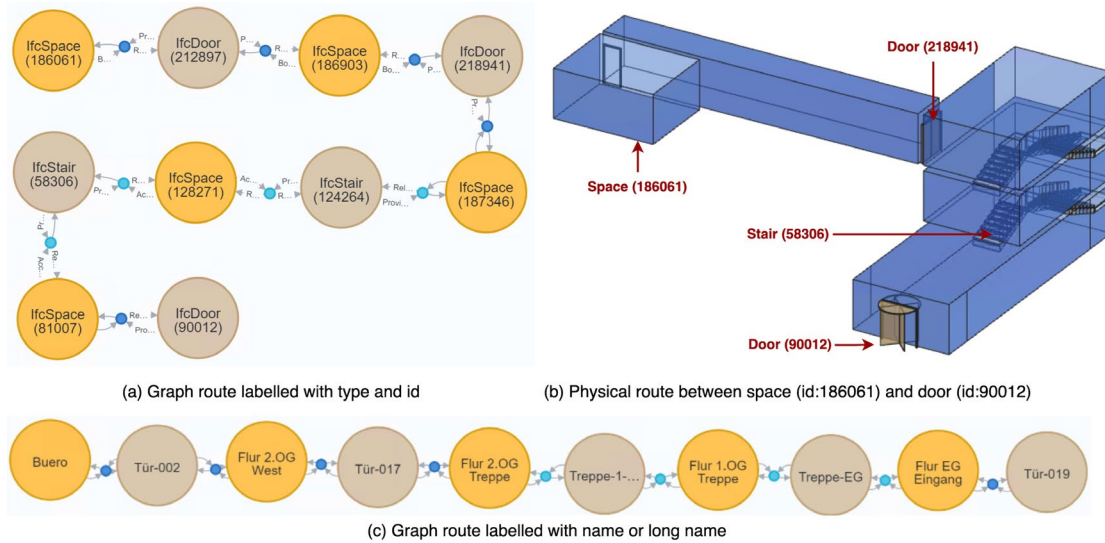


Figure 10 Path between space (id:186061) and door (id:90012), including (a) the graph route labelled with type and id, (b) the physical route, and (c) the graph route labelled with name or long name in German, where spaces are yellow nodes, stairs and doors are light brown nodes.

#### 4.3 Modified Dorm model

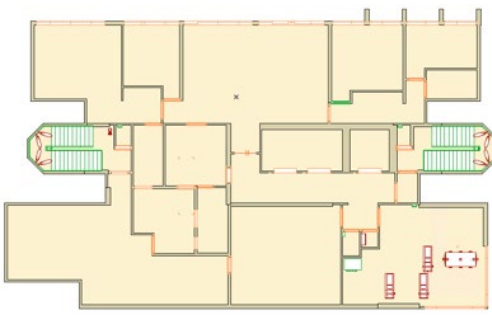
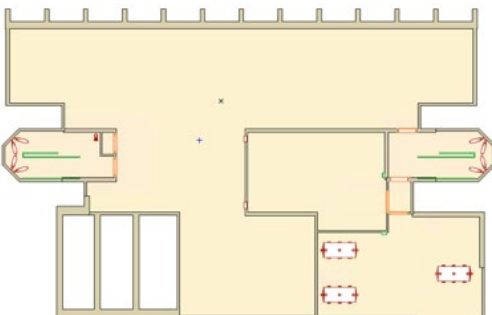
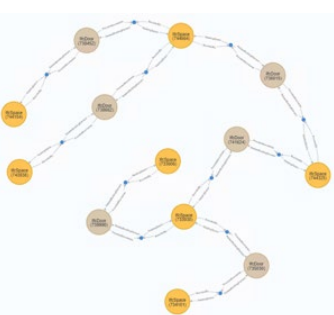
The modified Dorm building is used to further test if the proposed approach can work with a more complicated indoor environment. This 10-storey building includes 312 spaces in total, with each floor having a distinctive floor plan.

##### 4.3.1 Horizontal connectivity graphs for individual storeys

Through the query in Listing 1, which was completed in 6220 ms, the horizontal connectivity graphs for individual storeys were retrieved. Table 4 presents the floor plans and the queried connectivity graphs (spaces as yellow nodes, doors as light brown nodes) for 4 of the 10 floors, including Level 4, Level 6, Level 29, and the flat roof.

Table 4 Floor plans and corresponding connectivity graphs for the modified Dorm building.

| Floor | Floor Plan | Connectivity Graph |
|-------|------------|--------------------|
| 4/F   |            |                    |
| 6/F   |            |                    |

| Floor     | Floor Plan                                                                        | Connectivity Graph                                                                 |
|-----------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 29/F      |  |  |
| Flat Roof |  |  |

### 4.3.2 Vertical connectivity graph

Using the algorithm in Listing 2, the connectivity between storeys were identified, as presented in Figure 11. In the Dorm model, stairs do not have independent representations but are assemblies of their components, including member (IfcMember), stair flight (IfcStairFlight), slab (IfcSlab), and railing (IfcRailing). Figure 12 shows an example of stair assembly by using stair (662858), which include 2 stair flights, 1 railing, 1 slab, and 10 members. In this case, the stair flights were used to establish vertical connection.

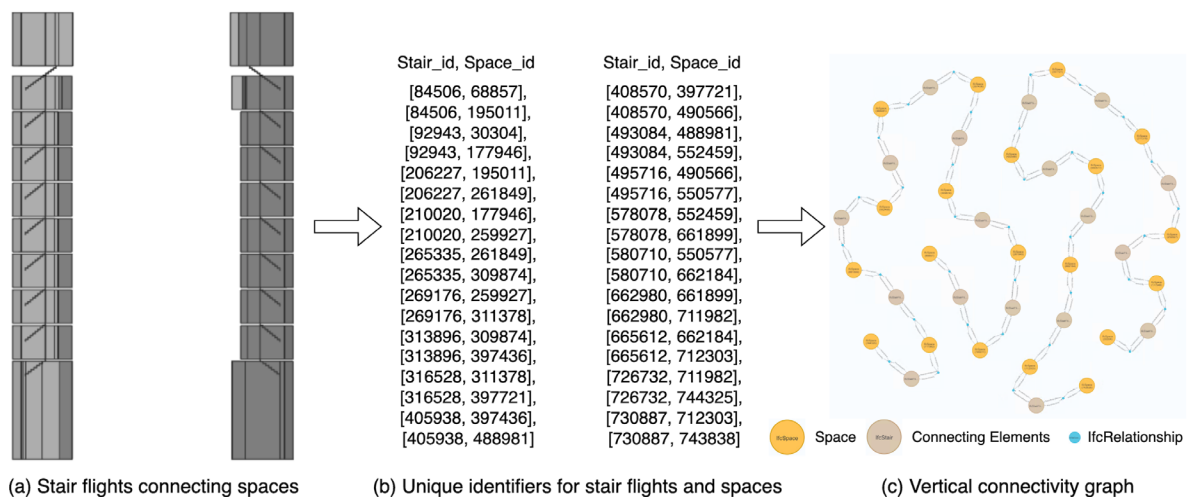


Figure 11 Stair flights connecting spaces.

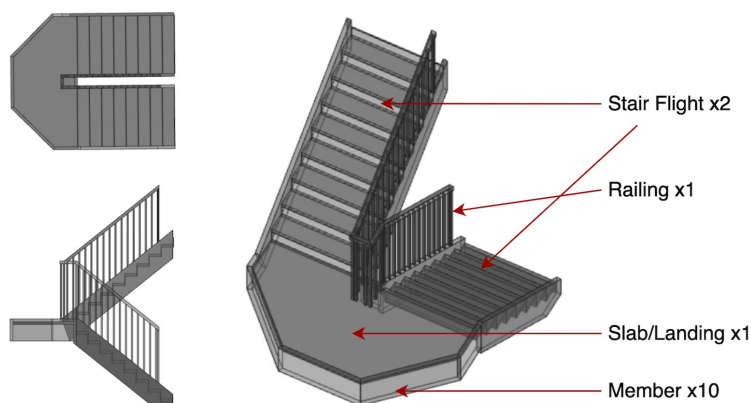


Figure 12 Stair (662858) comprised of 2 stair flights, 1 railing, 1 slab, and 10 members.

#### 4.3.3 Building-wide connectivity graph for the modified Dorm building

The building-wide connectivity graph (see Figure 13) was created by combining the vertical connectivity graph and the horizontal connectivity graph. This final graph includes 646 boundary relationships, 312 spaces, 296 doors, 30 virtual boundaries, 18 stair flights. With this building-wide connectivity graph, the shortest path between any two nodes can be queried. Two nodes from this graph were randomly selected for testing, one is a room in floor '5/F' with an id of 203340, another is also a room, in the flat roof with an id of 733606. The query in Listing 4 was used to find the shortest path between them, and the search was completed in 150 ms. The graph path and the physical path between these two spaces are presented in Figure 14.

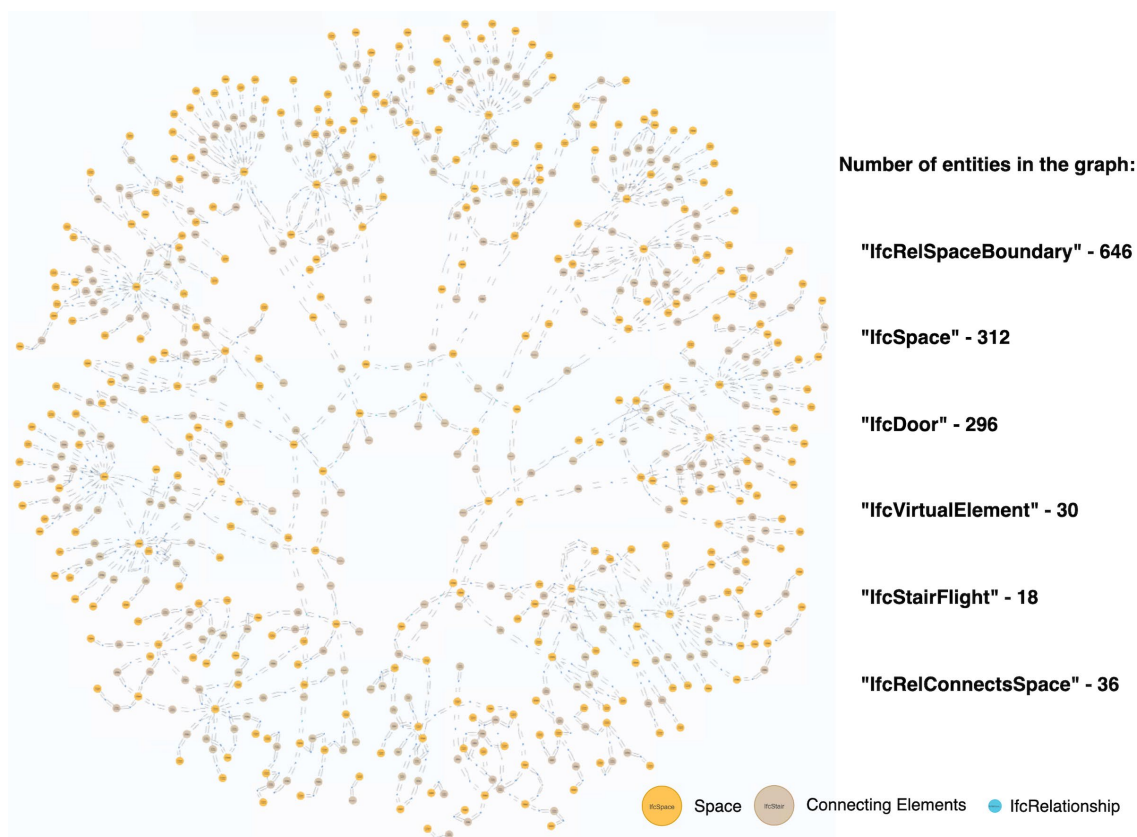


Figure 13 Building-wide connectivity graph for the modified Dorm building.

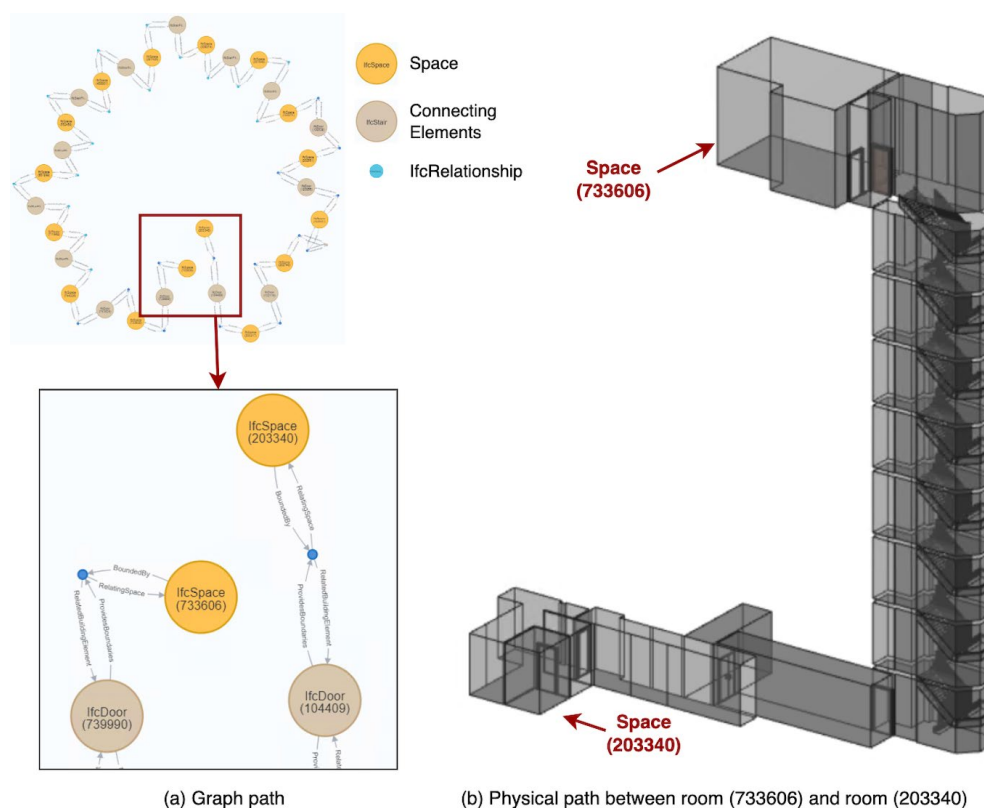


Figure 14 (a) Graph path and (b) physical path between room (733606) and room (203340).

## 5 Discussion

This paper proposes a new semantics-based approach for constructing connectivity graph that can be used in indoor pathfinding. The proposed approach is different from previous studies in three distinct ways, which are discussed in the following sections. (1) The construction of the connectivity graph. While the conventional geometry-based approach uses geometric information from 2D floor plans, this study mainly used semantic information from IFC through IFC-Graph, especially the relationship between space and space boundary. (2) Pathfinding strategy. Previous studies mainly used point-to-point pathfinding that requires accurate location coordinates, while this study enables object-to-object pathfinding, where accurate location coordinates are not mandatory. Objects here refer to any elements in the connectivity graph, such as spaces, doors, and facilities. (3) The visualisation of the path (or visual aid). Previous studies used geometric points and lines, while this study used the 3D (body) geometries of involved elements. These features make the proposed approach more efficient in generating navigable connectivity graph (by avoiding computing-intensive geometry processing) and can fit into digital twin applications by integrating with other technologies, such as augmented reality (AR).

### 5.1 Semantics-based object-to-object pathfinding Vs geometry-based point-to-point pathfinding

The semantics-based connectivity graphs enable object-to-object pathfinding based on the rich semantics provided by IFC. Instead of finding geometric points along the path, this approach finds meaningful objects. As presented above, the connectivity graph can be used to find the path between any two nodes (e.g., space-to-space or space-to-exit). This is different from the previous coordinates-based point-to-point pathfinding that uses geometric information and requires high accuracy indoor positioning systems.

First, the semantics-based object-to-object pathfinding has a low requirement on the accuracy of location (room-level) and would suit more with indoor pathfinding scenarios where the accuracy of the indoor positioning system is not that high. According to Taneja et al. [33], the indoor environments present a challenge to all types of positioning technologies and systems. While high-

accuracy indoor positioning can be achieved at additional cost, the general accuracy of WIFI-based indoor positioning systems is just between 2 m and 5 m [58]. This level of accuracy is not sufficient for most geometry-based approaches but works fine with the semantics-based approach. For the semantics-based approach, room-level accuracy is enough, as it just needs to know which space (e.g., room, corridor, or staircase) the user is in.

A second benefit of semantics-based pathfinding comes from the semantics being used, which can facilitate pathfinding. For example, the query in Listing 5 can be used to find the paths from a room (e.g., 203340) to all external doors (exit) by additionally using the 'InternalOrExternalBoundary' attribute of the IfcRelSpaceBoundary relationship. External doors are those connected to IfcRelSpaceBoundary nodes that have a value of 'EXTERNAL' for the 'InternalOrExternalBoundary' attribute.

Listing 5 Graph query for finding the paths between room (203340) and all external doors (exit).

```
match p = shortestPath((n1{id:203340})-[*]-(n2:IfcDoor))
where (n2)--(:IfcRelSpaceBoundary {InternalOrExternalBoundary:'EXTERNAL'})
return p
```

This query in Listing 5 on the modified Dorm model was completed in 339 ms, and Figure 15 presents the queried paths between room (203340) and all the 4 external doors (Door1 to Door4) on the '4/F' floor, which is the actual ground floor of the Dorm building. The shortest path to exit is through Door1, which is next to the stair.

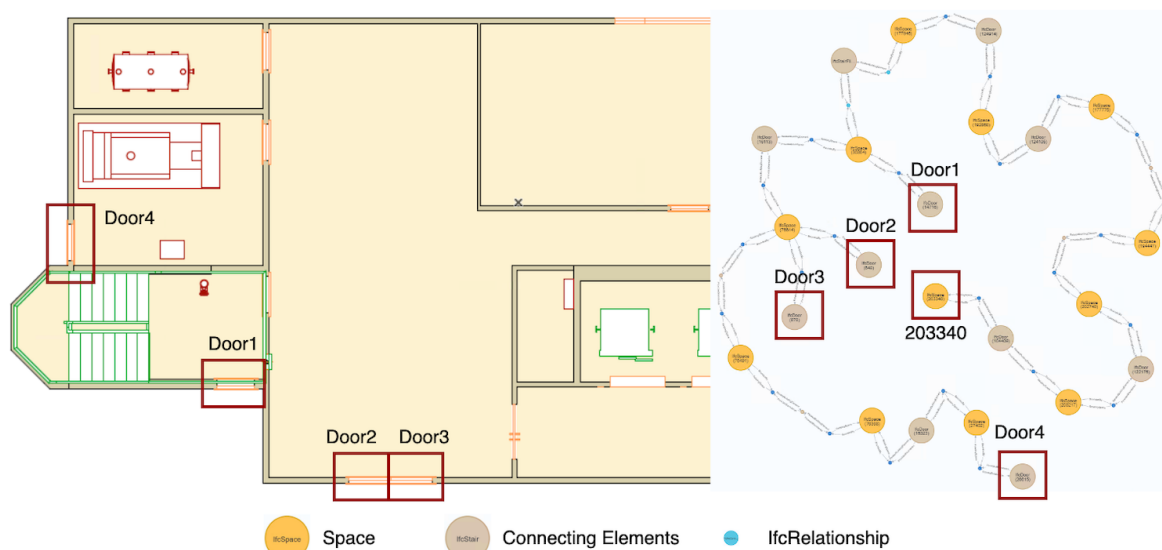


Figure 15 Room (203340) to the four external doors.

It is also even possible to add a filter in the 'where' clause using the attributes of doors to limit the size of doors in the returned path, which would help in some situations where there is a requirement on the size of doors in the path. In addition, this study distinguishes doors that connect the same two spaces. A link is created for each door. This is different from the connectivity graph created by Yang et al. [59], where despite the number of openings between two spaces, only one link was created. The differentiation of doors would help in situations where the doors have different features, e.g., size. All semantics defined in IFC can be directly used to define constraints in a query, making the semantics-based approach flexible and versatile for more specific pathfinding use cases. This is supported by the IFC data model that is implemented in IFC-Graph. However, it is also possible to introduce information not defined in IFC, like the size of space, into the connectivity graph, in that case, additional geometry processing is required. Figure 16 shows the screenshots of the neo4j databases, which are examples of the data model for space, door, and the space boundary relationship. The list of IFC entities above each subfigure are labels showing the entities to which the

instance belongs. For example, Figure 16 (a) shows that the space instance (id:177561) is an instance of any of IfcObject, IfcObjectDefinition, IfcProduct, IfcRoot, IfcSpace, IfcSpatialElement, and IfcSpatialStructureElement. These labels are created during IFC-Graph generation and are essential for information query.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                     |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------|------|------|-----------------|---------|----------|-------------------------|----------------------------|----------|------|----------|---------------------------|----------|----|--------|------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----|------|-----|-----------------|------------------------|----------|-------------------------------------|---------------|----------|--------------|-------|----------------|-------|-----|--------|------|----------|------|---------|
| <div> <div>IfcObject</div> <div>IfcObjectDefinition</div> <div>IfcProduct</div> <div>IfcRoot</div> <div>IfcSpace</div> <div>IfcSpatialElement</div> <div>IfcSpatialStructureElement</div> </div> <table> <tr><td>&lt;elementId&gt;</td><td>194</td></tr> <tr><td>&lt;id&gt;</td><td>194</td></tr> <tr><td>CompositionType</td><td>ELEMENT</td></tr> <tr><td>GlobalId</td><td>2jluF41IDCvQC6DkPEc4Ou</td></tr> <tr><td>LongName</td><td>Room 501</td></tr> <tr><td>Name</td><td>2307</td></tr> <tr><td>PredefinedType</td><td>SPACE</td></tr> <tr><td>id</td><td>177561</td></tr> <tr><td>name</td><td>IfcSpace</td></tr> </table> <p>(a) Space - Room</p> | <elementId>                         | 194  | <id> | 194  | CompositionType | ELEMENT | GlobalId | 2jluF41IDCvQC6DkPEc4Ou  | LongName                   | Room 501 | Name | 2307     | PredefinedType            | SPACE    | id | 177561 | name | IfcSpace            | <div> <div>IfcObject</div> <div>IfcObjectDefinition</div> <div>IfcProduct</div> <div>IfcRoot</div> <div>IfcSpace</div> <div>IfcSpatialElement</div> <div>IfcSpatialStructureElement</div> </div> <table> <tr><td>&lt;elementId&gt;</td><td>173</td></tr> <tr><td>&lt;id&gt;</td><td>173</td></tr> <tr><td>CompositionType</td><td>ELEMENT</td></tr> <tr><td>GlobalId</td><td>2jluF41IDCvQC6DkPEc4O_</td></tr> <tr><td>LongName</td><td>Corridor</td></tr> <tr><td>Name</td><td>2309</td></tr> <tr><td>PredefinedType</td><td>SPACE</td></tr> <tr><td>id</td><td>177927</td></tr> <tr><td>name</td><td>IfcSpace</td></tr> </table> <p>(b) Space - Corridor</p>                               | <elementId> | 173 | <id> | 173 | CompositionType | ELEMENT                | GlobalId | 2jluF41IDCvQC6DkPEc4O_              | LongName      | Corridor | Name         | 2309  | PredefinedType | SPACE | id  | 177927 | name | IfcSpace |      |         |
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| LongName                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Room 501                            |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
| Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2307                                |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
| PredefinedType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SPACE                               |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
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| LongName                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Corridor                            |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
| Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2309                                |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
| PredefinedType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SPACE                               |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
| id                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 177927                              |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
| name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | IfcSpace                            |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
| <div> <div>IfcRelConnects</div> <div>IfcRelSpaceBoundary</div> <div>IfcRelationship</div> <div>IfcRoot</div> </div> <table> <tr><td>&lt;elementId&gt;</td><td>1228</td></tr> <tr><td>&lt;id&gt;</td><td>1228</td></tr> <tr><td>Description</td><td>2a</td></tr> <tr><td>GlobalId</td><td>2PaR3koBjIXe1XcQ6ZuE\$9</td></tr> <tr><td>InternalOrExternalBoundary</td><td>INTERNAL</td></tr> <tr><td>Name</td><td>2ndLevel</td></tr> <tr><td>PhysicalOrVirtualBoundary</td><td>PHYSICAL</td></tr> <tr><td>id</td><td>888885</td></tr> <tr><td>name</td><td>IfcRelSpaceBoundary</td></tr> </table> <p>(c) Space boundary</p>                                   | <elementId>                         | 1228 | <id> | 1228 | Description     | 2a      | GlobalId | 2PaR3koBjIXe1XcQ6ZuE\$9 | InternalOrExternalBoundary | INTERNAL | Name | 2ndLevel | PhysicalOrVirtualBoundary | PHYSICAL | id | 888885 | name | IfcRelSpaceBoundary | <div> <div>IfcBuildingElement</div> <div>IfcDoor</div> <div>IfcElement</div> <div>IfcObject</div> <div>IfcObjectDefinition</div> <div>IfcProduct</div> <div>IfcRoot</div> </div> <table> <tr><td>&lt;elementId&gt;</td><td>7</td></tr> <tr><td>&lt;id&gt;</td><td>7</td></tr> <tr><td>GlobalId</td><td>2ejTfxI399fghzDm31xBzj</td></tr> <tr><td>Name</td><td>M_Single-Flush:0915 x 2134mm:610546</td></tr> <tr><td>OverallHeight</td><td>2134.0</td></tr> <tr><td>OverallWidth</td><td>915.0</td></tr> <tr><td>PredefinedType</td><td>DOOR</td></tr> <tr><td>Tag</td><td>610546</td></tr> <tr><td>id</td><td>1441</td></tr> <tr><td>name</td><td>IfcDoor</td></tr> </table> <p>(d) Door</p> | <elementId> | 7   | <id> | 7   | GlobalId        | 2ejTfxI399fghzDm31xBzj | Name     | M_Single-Flush:0915 x 2134mm:610546 | OverallHeight | 2134.0   | OverallWidth | 915.0 | PredefinedType | DOOR  | Tag | 610546 | id   | 1441     | name | IfcDoor |
| <elementId>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1228                                |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
| <id>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1228                                |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
| Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2a                                  |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
| GlobalId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2PaR3koBjIXe1XcQ6ZuE\$9             |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
| InternalOrExternalBoundary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | INTERNAL                            |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
| Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2ndLevel                            |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
| PhysicalOrVirtualBoundary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | PHYSICAL                            |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
| id                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 888885                              |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
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| <elementId>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 7                                   |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
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| GlobalId                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2ejTfxI399fghzDm31xBzj              |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
| Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | M_Single-Flush:0915 x 2134mm:610546 |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
| OverallHeight                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2134.0                              |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
| OverallWidth                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 915.0                               |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
| PredefinedType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | DOOR                                |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
| Tag                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 610546                              |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
| id                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1441                                |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |
| name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | IfcDoor                             |      |      |      |                 |         |          |                         |                            |          |      |          |                           |          |    |        |      |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |     |      |     |                 |                        |          |                                     |               |          |              |       |                |       |     |        |      |          |      |         |

Figure 16 Data model examples for (a) space-room, (b) space-corridor, (c) space boundary, and (d) door.

## 5.2 Flexible data structure for various pathfinding use cases

The semantics-based approach is based on the flexible graph structure, which can be easily extended. New semantic information can be added to the graph to meet additional requirements. In the previous section, examples for space-to-space and space-to-exit pathfinding have been provided, which only used information from the initial navigation data model. There can be additional needs, for example, to find the available equipment or facilities near the user, such as fire suppression terminals. To meet this need, semantic information from two IFC entities, i.e., IfcRelContainedInSpatialStructure and IfcFireSuppressionTerminal, can be added to the building-wide connectivity graph. Once the connectivity graph has been updated with the required information, queries can be designed to find the paths. Listing 6(a) is the query for finding all the fire suppression terminals near room (203340), while Listing 6(b) is the query for finding the path between room (203340) and the nearest fire suppression terminal (193956).

Listing 6 Query for finding fire suppression terminals near room (203340).

|     |                                                                                               |
|-----|-----------------------------------------------------------------------------------------------|
| (a) | <pre>match p = shortestPath((n1{id:203340})-[*]-(:IfcFireSuppressionTerminal)) return p</pre> |
| (b) | <pre>match p = shortestPath((n1{id:203340})-[*]-(n2{id:193956})) return p</pre>               |

Using the Dorm model, the first query was completed in 1720 ms, and the second query was completed in 49 ms. The queried paths are presented in Figure 17, including the paths between room (203340) and all nearby fire suppression terminals (Figure 17(a)), the path between room (203340) and

the nearest fire suppression terminal (193956) (see Figure 17(a)), as well as a 2D view of the physical path in Figure 17(c) and a 3D view in Figure 17(d). This semantics-based approach can utilise the rich semantic information in IFC to answer questions like ‘where is the nearest X’, which is difficult for the conventional geometry-based approach to answer. Other objects can be added to the connectivity graph when needed.

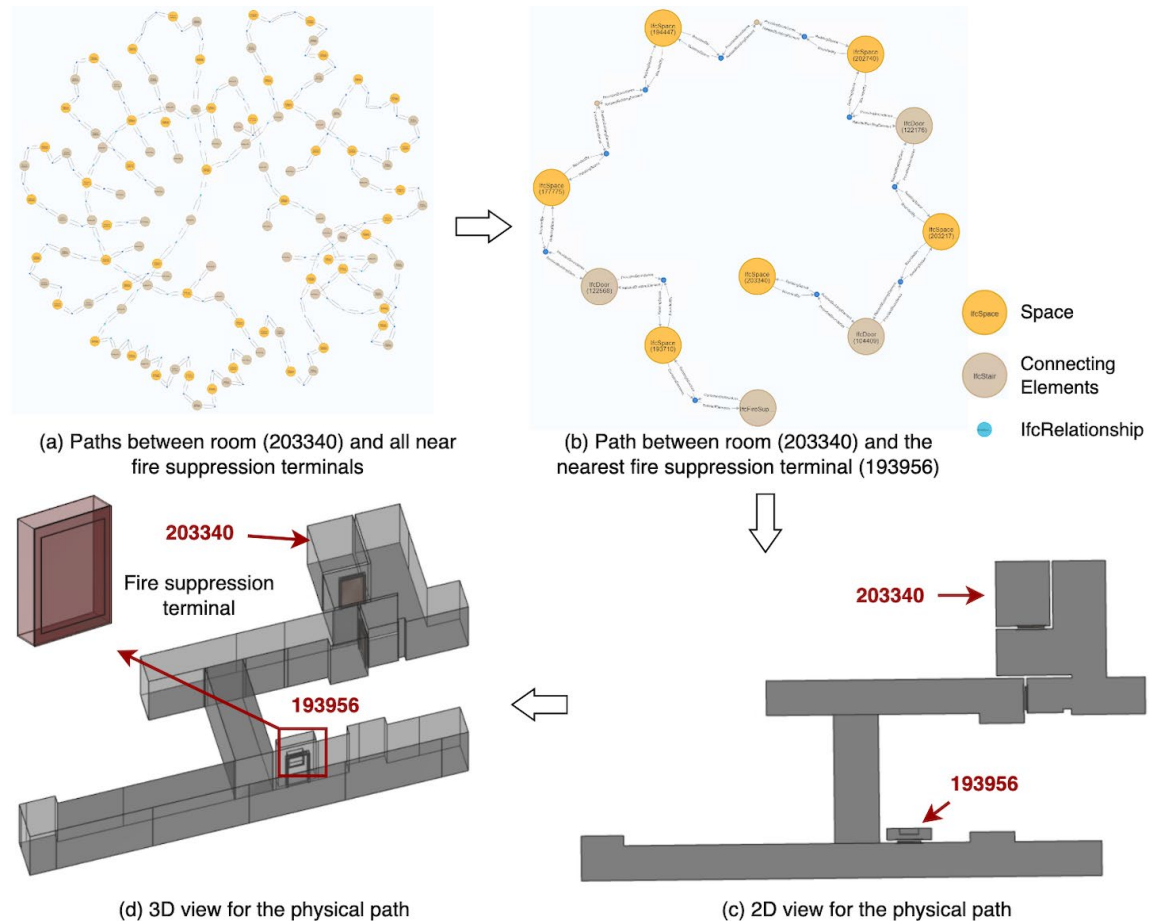


Figure 17 Room (203340) to the nearest fire suppression terminal (193956).

### 5.3 Efficiency of the semantics-based approach

Table 5 shows the time consumed by each step for the construction of the navigation graphs for the Institute model and the modified Dorm model. The construction of the navigation graphs for the Institute model and the modified Dorm model was completed in 6.72 seconds and 18.97 seconds, respectively. The measured time is wall time, which is conservative, because it includes the active computation time and waiting time. The efficiency can be further improved by about 10% if the vertical connectivity had been defined in IFC, as no time cost is needed for the generation of the vertical connectivity graph.

Table 5 Time for constructing the semantics-based navigation graph.

|   |                                                     | Time (seconds) |               |
|---|-----------------------------------------------------|----------------|---------------|
|   |                                                     | Institute      | Modified Dorm |
| 1 | Horizontal connectivity graph query                 | 2.43           | 6.22          |
| 2 | Horizontal connectivity graph generation            | 2.82           | 8.02          |
| 3 | Vertical connectivity graph generation              | 0.80           | 2.00          |
| 4 | Merging horizontal and vertical connectivity graphs | 0.67           | 2.73          |
|   | Total time for the construction of the graph        | 6.72           | 18.97         |

It is difficult to do a comprehensive comparison with other approaches due to the differences in the building models and hardware used. Below for reference are brief summaries for several previous studies on their time consumed for the construction of navigation networks and pathfinding.

Table 6 Time consumed in previous studies for the construction of navigation networks and shortest pathfinding.

| Study               | Year | Summary                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wong et al. [5]     | 2023 | <ul style="list-style-type: none"> <li>The visibility graph method was applied to buildings with 3 to 40 floors.</li> <li>The time for pathfinding on a phone is between 0.06 second and 2.95 seconds depending on the number of stories, using a new two-stage pathfinding method.</li> <li>The time for pathfinding is between 0.05 second and 21.13 seconds using the conventional Dijkstra's algorithm.</li> </ul> |
| Man et al. [16]     | 2017 | <ul style="list-style-type: none"> <li>A grid-based approach and a spatial subdivision method were applied to two buildings with 2 to 3 floors.</li> <li>The calculation time for the grid-based approach was between 0.002 second to 9.022 seconds, depending on the size of the grid.</li> </ul>                                                                                                                     |
| Teo et al. [9]      | 2016 | <ul style="list-style-type: none"> <li>A multi-purpose geometric network model (MGNM) was proposed for indoor and outdoor combined route planning, applied to a building with 4 floors.</li> <li>It took 63 seconds to construct the proposed network.</li> <li>Time for pathfinding was between 8.55 seconds to 67.79 seconds, depending on the number of transition points.</li> </ul>                               |
| Lin et al. [24]     | 2013 | <ul style="list-style-type: none"> <li>A grid-based approach, together with Fast Marching Method (FMM), was applied to the second floor of a five-storey building.</li> <li>It took 3 seconds for the discretization and mapping of the space with a 500x500 grid, and 13.87 seconds for pathfinding.</li> </ul>                                                                                                       |
| The proposed method | 2024 | <ul style="list-style-type: none"> <li>A semantics-based approach applied to buildings with 5 and 10 floors.</li> <li>The time for constructing the connectivity graph is between 6.72 seconds to 18.97 seconds</li> <li>The time for pathfinding is between 0.04 second to 0.15 second.</li> </ul>                                                                                                                    |

#### 5.4 Semantics-based pathfinding/navigation and digital twin (DT) technology

This approach can be best used with digital twin (DT) technology equipped with indoor positioning system, sensor network, Internet of Things (IoT), and augmented reality (AR). Indoor positioning system, e.g., WIFI-based, can be used to get the rough location of the user within the building. The rough location can be used to find out which space the user is in and further get the space id, which can be used as the start node. Once the destination is determined by semantics search, a path can be queried from the connectivity graph. For paths with the same length, the selection is up to the user. Once a path is selected, navigation can be started through AR. During navigation, the real-time location of the user will be tracked and updated in the system and compared against the connectivity graph. Accordingly, the current space and the next space/objects in the path will be highlighted as visual aid to guide the movement of the user from one space to the next. For large and angled spaces, transparent colours will be used for walls, so the users can see the full boundaries of the spaces. If the user is deviated from the planned route, a new path will be calculated using the new location. In the meanwhile, real-time data from sensor network will be transferred to the digital model via IoT and used to update the status of the building, e.g., the status of doors, the location of obstacles [42], etc. The status of the building can be used to adjust the route in an ad hoc way.

In this context, BIM/GIS integration will be needed as well. BIM is mainly a tool for collating information, lacking the capability of information analysis, while GIS is a tool for information management and analysis, including network analysis for pathfinding [60]. According to Teo and Cho [9], BIM/GIS integration for emergency response is a priority in smart city applications. It is,

therefore, common to see studies that convert BIM data, e.g., IFC, into GIS data (raster data or vector data in shapefile or geodatabase) for performing pathfinding, especially in the context of indoor/outdoor integrated navigation. For example, Teo and Cho [9] extracted building information from IFC and use it to generate an indoor network graph, which was stored in a geodatabase. Similar studies include Lin et al. [12,61] and Isikdag et al. [34] that stored navigation information in a GIS format. In this regard, problems in BIM/GIS integration should be concerned, such as geometry conversion, geo-referencing [62,63], and class mapping.

### 5.5 'Shortest' path in semantics-based indoor pathfinding

The use of the number of elements involved in a path as the measure of the length might be the most questionable part for the semantics-based indoor pathfinding. This understanding probably comes from people's experiencing with outdoor pathfinding, e.g., using Google Map, that uses either distance or time as the measure. In an indoor environment, according to Liu [14] the shortest-distance path is not as important as those for outdoor environments, especially in a non-emergency scenario, and it is acceptable to use the number of involved elements as the measure of distance in the indoor environment. This is mainly because of the following reasons.

Indoor pathfinding is different from outdoor pathfinding in many ways, including the structure of the connectivity graph. The connectivity graph for outdoor is usually a complex network structure composed of path segments. There can be many paths connecting two points within the outdoor network. Figure 18 shows the possible paths between point A and point B with a length of 4 in a 3-by-3 network. There will be more paths when there is no limit on the length of the path. In addition, the length of path segments varies, from meters to kilometres. For example, the shortest road segment in the Strategic Route Network of United Kingdom is 6 metres, while the longest is 7436 metres [64]. These features of outdoor network require the use of physical distance or time as the measure of 'distance', not the number of road segments.

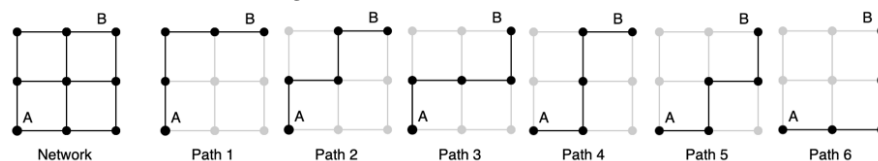


Figure 18 Paths between Point A and Point B in a 3-by-3 network structure.

In contrast, the navigation connectivity graphs for indoor environments usually have a simpler form such as the tree structure (e.g., for the Institute model) or cycle structure or a combination of both, as shown in Figure 19. Because of this, compared with the outdoor paths, there are fewer options for indoor paths [14]. For a tree structure, any two nodes in the graph are connected by one and only one path (see Figure 20(a)). There will be a few more options when there are cycles (see Figure 20(b)). The relatively simple layouts of buildings come from architects who aim to design buildings providing easy access [2,65]. Complex layouts adversely affect users' experience.

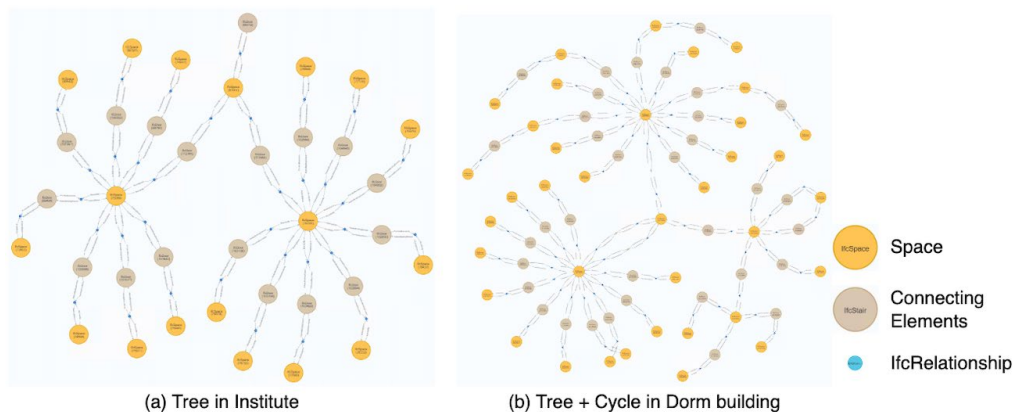


Figure 19 Tree structure in the Institute model and the tree + cycle structure in Dorm building model.

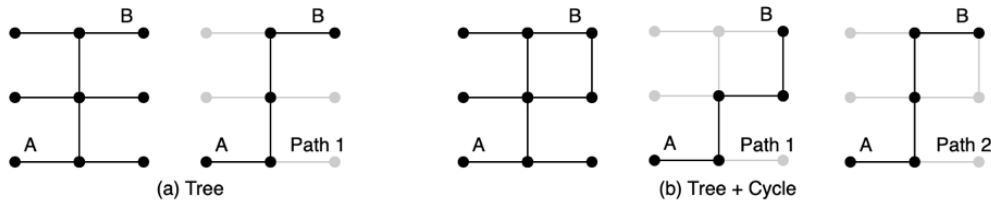


Figure 20 Paths between A and B in (a) tree structure and (b) tree + cycle structure.

Due to the features of indoor 'network', in many cases, for example in the Institute model, the least-element paths are found to also be the shortest-distance paths. This can be attributed to the fact that in indoor environment, users must move through a series of key circulation elements (such as corridors, doors, or stairs) to reach the destination. Even for multistorey buildings, an emergent space-to-exit pathfinding is still mainly about finding the nearest stair on the same floor, despite the number of floors, as links between storeys are fixed [5]. Nevertheless, the relationship between these two types of paths needs to be further investigated. It is worth noting that geometric information is available within the IFC-Graph, so edges and nodes can be labelled with time or distance attributes, just like the study by Khalili and Chua [40].

### 5.6 Limitations and future work

For semantics-based pathfinding, the following aspects can be further improved.

(1) Combining semantics with geometry in indoor pathfinding. This paper currently focuses on the most fundamental connectivity graph for general indoor pathfinding, where physical distance is not considered. So, it can now be directly applied in non-emergency situations, such as navigating through a university. However, in different situation, the selection of the optimal path depends on multiple factors, including physical distance. For example, physical distance may be preferred by some applications, such as emergency response and compliance check. In this regard, methods should be developed to embed distance information into the semantics-based connectivity graph, e.g., by adding a weight attribute to nodes or by properly subdividing indoor space, as suggested by Liu [14], using virtual boundary (IfcVirtualElement) or function boundary [59]. A preliminary experiment has been conducted and presented in Figure 21.

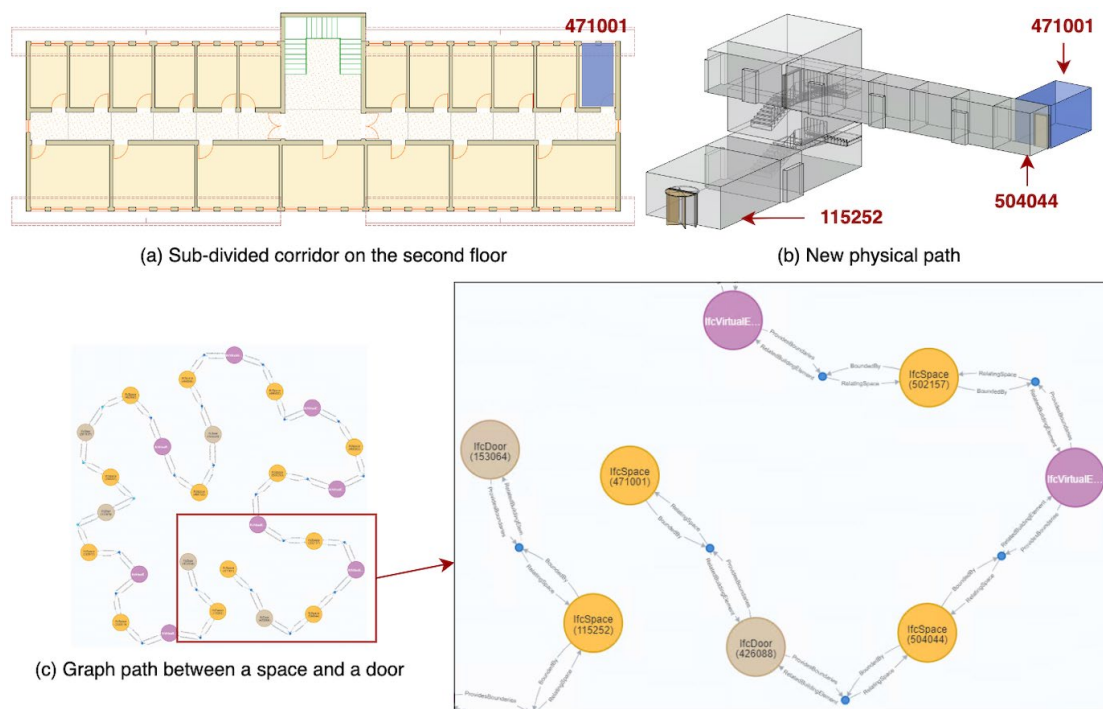


Figure 21 Pathfinding based on subdivided space. (a) The sub-divided corridor for the second floor, (b) the new physical path, and (c) the graph path between space (471001) and door (153064).

In this preliminary experiment, the corridors in the Institute model were subdivided into equal sub-spaces. An example of the divided corridors is given in Figure 21(a) using the second floor. After constructing the semantics-based navigation graph, the path between a space (471001) and a door (153064) was queried, which is presented in Figure 21(b) and Figure 21(c). From Figure 21 (c), it can be observed that, the subspaces are connected by IfcVirtualElement. This experiment shows that it is possible to embed distance information into graph by subdividing space. However, a more comprehensive work in this regard should be carried out in the future.

(2) Enabling pathfinding for multiple navigation modes: walking, flying, rolling. This paper is currently targeting on human user (walking); additional navigation modes should be included in the future, such as flying for drones or rolling for wheel-based robots. In this regard, additional information should be added to the graph. For example, to enable the flying mode, information about windows (both internal and external) would be added, which can be queried using the IfcRelSpaceBoundary relationship. The query in Listing 7 can then be used to find the paths from a room (203340) to the near external windows.

Listing 7 Graph query for finding the paths between room (203340) and external windows.

```
match p = shortestPath((n1{id:203340})-[*]-(n2:IfcWindow))
where (n2)--(:IfcRelSpaceBoundary {InternalOrExternalBoundary:'EXTERNAL'})
return p
```

(3) Integrating the semantics-based connectivity graph with other modules for indoor navigation. This paper proposes an indoor navigation model and prove its feasibility. However, to use it in practice, the navigation model should be integrated with other modules, such as indoor localization and indoor guidance instruction. In addition, a better visualization strategy for the physical path should be designed to enhance human-system interaction.

## 6 Conclusions

This paper proposed a new semantics-based approach for indoor pathfinding by constructing connectivity graph using IFC-Graph. This approach is different from previous studies that mainly used geometric information from either CAD or BIM. This new approach tries to avoid the use of computing-intensive geometry processing and uses the rich semantic information in IFC data instead. The proposed methods were validated using two real-world building models. Use cases have been provided for space-to-space, space-to-exit, and space-to-facility pathfinding. The main findings and outputs are as follows.

(1) As a combination of geometric, semantic, and topological information, IFC serves as a good source for constructing indoor navigation connectivity graph. However, it currently does not provide semantic information about vertical connection between storeys. In this paper, a customised relationship entity, IfcRelConnectsSpace, was developed for this purpose. (2) The semantics-based approach is different from the previous geometry-based approach, in terms of the construction of the navigable connectivity graph, pathfinding pattern, and the visualisation of the path. In this paper, the majority of the connectivity graph was established by querying IFC-Graph, which avoids computing-intensive geometry processing and builds the connectivity graph in seconds. (3) The semantics-based approach uses semantic information from IFC, which enables object-to-object pathfinding, which is difficult for the conventional geometry-based approach. It is also highly flexible to add semantic information to empower various tailored scenarios, compared to the conventional geometry-based approach. (4) The object-to-object pathfinding can enable additional pathfinding applications, for example, finding paths to all external doors for egress or paths to the near available equipment of interest. The semantics-based approach does not mean a semantics-only approach, geometric information was used for establishing the vertical links between storeys and the visualisation of the queried path, and the use of 3D shapes of elements as visual aids is intuitive for human users.

While the semantics-based connectivity graph has demonstrated its potential and flexibility for indoor pathfinding, additional future work should be carried out to make it more robust and to

support more applications. For example, it can be combined with the geometry-based approach to enable shortest-distance pathfinding, which is necessary for certain scenarios in compliance check, e.g., calculating the escape distance from a specific room to any exit door. Finally, the navigation model should be extended to support more navigation modes, such as flying and rolling.

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**Author Contributions:**

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